

More on Friston's Free Energy Principle

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method: auto_caption

attention we my application to oist has gotten me an interview at the end of may i'll talk to some professors in okinawa on zoom so for this week's video i'm gonna practice my pitch a few weeks ago my talking squid video essay was about fristen's free energy principle neuroscience's latest model of the sentient being i described a project involving bees i i still like the idea but it's another classic example of me getting way too complicated way too quick the project i actually proposed to oist is much more simple carl fristen demonstrated his free energy principle by training a program to navigate a maze i want to train two programs to navigate a maze one predator and one prey the predator wants to minimize its distance to the prey the prey wants to maximize its distance from the predator at hoist i'd like to make this work continuously but for now let's leave it discreet like a board game in each time step the predator and prey both move one space up left down or right the two agents compete learning how to best interpret their perceptions of reality by minimizing free energy in a generative adversarial network last time i talked about free energy i skipped most of the deeper math now i've read and reread enough tutorials to go over one step by step explaining how it applies to my project here also thank you professor giuntani for recommending this tutorial from medium.com reality can be called a collection of hidden variables which i'll label s meaning state our reality is big and complicated but the predator and prey have a pretty simple reality we can edit in a photo editor every white tile is open every black tile is an obstacle let's call this area an x by y matrix of ones and zeros let's add two layers which are just zeros except one one at the predator's position and another one at the prey position this reality is a three by x by y matrix of ones and zeros but all this is hidden no one gets to see every part of reality at once except us deities who made this reality in a photo editor and prey only observe reality within a few spaces of their position let's say z spaces so predator and prey get 3 by z by z matrices of ones and zeros which i'll label o meaning observation they both use these small observations to understand their larger reality generating an inferred state given some observation what's the probability reality is one state or another if reality was one state or the

other what's the probability to get this observation this is called active inference our conceptualization of reality should maximize the likelihood of the observations we're getting maximizing the probability of our observations is the same as minimizing the negative log of the probability of our also called our surprise okay the tutorial notes that the probability of an event is equal to the sum of the probabilities of that event occurring alongside every alternative for another event for instance uh the probability of rolling a six on a die is one out of six and if we rolled another die next to it we can think of that one-sixth probability as divided into sections based on the other die so we can include all possibilities for reality inside this expression we're calling surprise we also multiply by q of s divided by a q of s whatever q of s is q of s divided by q of s is one and multiplying by one is a perfectly fine thing to do now we're trying to minimize this surprise and that's hard it would be a little easier to minimize this thing because this q of s is outside the logarithm this expression is called free energy thankfully it just so happens that free energy is the upper bound of surprise so minimizing free energy does minimize surprise approximately the tutorial loses this minus sign by inverting this fraction in the logarithm then the tutorial shows another way to represent free energy with a notation i had never heard of before called leibler divergence also called relative entropy the kl of two probability distributions is a measure of difference which best measures zero when the probability distributions are exactly the same if we can manage that here then the free energy happens to be equal to the and we can probably get pretty close because q of s is an arbitrary nonsense function we made up we just have to nudge it closer to p of s given o to turn math into a metaphor we're making our private notion of the universe into something similar to what our observations are suggesting professor jun tani might suggest our private notion of the universe should involve long short-term memory networks okay i'm going to quote directly from the tutorial just to make sure i'm reading it right by minimizing free energy the arbitrary distribution q of s gets closer to the posterior p of s given o and if they match kl term becomes zero and free energy is exactly equal to surprise so the more we minimize free energy by wiggling q of s the more it becomes closer to since it's an upper bound and by wiggling parameters of p of s and s we can minimize surprise even further so that's what it means to actively infer for one discrete moment but because reality changes in every moment we can't stick to one discrete moment our inner model might not work so well with whatever we observe next so we have to improve our model in every time step and reality can change based on what we chose to do lately to minimize free energy we've got to choose our own actions responsibly at every time step we approximate these terms for every possible policy

we might like to consider our policies and actions several steps into the future but we can't predict the future so we've got to approximate it i'm skipping a few complicated steps here but the tutorial's final way to write free energy is like this the sum of cost and ambiguity to quote the tutorial again because i think this is the sort of thing i might be learning to understand for months or years the left term called cost or risk favors policies that will result in observations we like and the right term ambiguity quantifies how uncertain is the mapping between state and observations let's cover the tutorial's incredibly simple example and then try to stretch that example out to my predator slash prey project proposal suppose the universe has only two possible states your stomach has food or your stomach does not have food there are only two possible observations two feeling fed or feeling hungry there are also only two possible actions eating or not eating where does a mind fit into this squint or you'll miss says the tutorial we like to be fed and not hungry so we assign a higher to observation one fed in other words we express preferences over observations as probability p of o doesn't that sound a little backward i thought we'd be estimating probability but the concept of assigning probability to our observations makes it sound the other way around we want to feel fed so we're putting a higher probability on the chance of us feeling fed when we make a decision we consider our high likelihood to feel fed and that might inspire us to feed ourselves because that would explain the feeling fed to put it the tutorial's way we start by estimating the hidden states under each policy and end up acting such as we observe what we expect no wonder motivational speakers are all about optimism and vision the numbers say you go where you aim not everyone who thinks they become president becomes president but no one who thinks they won't become president becomes president they don't bother taking any of the steps so if we want our computer to do something for us we should tell it this is going to be accomplished i'm 100% sure do it so let's stretch this out to the and pray the reality here is bigger than two states food or no food there's a whole three by x by y matrix of zeros and ones the observations are bigger than boolean ii the predator and prey both have a three by z by z matrix of zeros and ones and they can tell how far apart they are from the other agent pred and prey both have four possible actions up left down or right i want the predator to chase the prey so i tell it closeness has high when the predator makes a decision it picks the action which leads to what i told it to expect at the same time i want the prey to escape from the predator so i tell it distance has a high probability when the prey makes a decision it picks to expect this predator and prey are animated by me hi but i'm confident with the right setup Fristen's free energy principle could get these two agents moving as if by their own thought how no matter

how much i read i don't think i'll ever find a tutorial which explains where the thoughts come from really there are no quote-unquote thoughts here when i made those generative adversarial networks which produced pokemon pictures i never wondered gosh where did that pokemon come from i never gave the generator a pokemon how did it get one the pokemon i see don't live in the generator or the picture the pokemon i see come within me the viewer the observer the beholder i've been reading a book called you are the universe by depak chopra the book tries to disprove a determinist notion of the mind it says the mind isn't limited by the brain because the brain is not the origin of the mind as the sort of math squid who makes this sort of video essay i'll always be ranting about the complexity of machine learning algorithms deep as the human brain but when i perceive the predator and pray seeming to make their own decisions even though i know their thoughts i'm seeing are my own i wonder where my thoughts are coming who does the thinking here at thinkster the tutorial gets more explicit with the math in another post and i'll get there eventually i'll get there eventually because i believe it's likely that i will get there eventually and therefore i reliably choose actions which lead me closer to getting there eventually because that's what i expect somehow there is little doubt in my mind that this is how the mind works i'll see you a few time steps in the future bye-bye by the way i've got a patreon at patreon.com/thinkster i want to thank all these squidlings and elder squids thanks [Music]